entropy splat forward through the ensuing stages of Figure 13.8. But, as it's currently conceived, the cyclic model offers no explanation of how or why the universe finds itself in the necessary configuration of Figure 13.8. Why, for instance, do six dimensions curl themselves up into a particular Calabi-Yau shape while one of the extra dimensions dutifully takes the shape of a spatial segment separating two three-branes? How is it that the two end-of-the -world three-branes line up so perfectly and attract each other with just the right force so that the stages in Figure 13.8 proceed as we've described? And, of critical importance, what actually happens when the two three-branes collide in the cyclic model's version of a bang?

On this last question, there is hope that the cyclic model's splat is less problematic than the singularity encountered at time zero in inflationary cosmology. Instead of all of space being infinitely compressed, in the cyclic approach only the single dimension between the branes gets squeezed down; the branes themselves experience overall expansion, not contraction, during each cycle. And this, Steinhardt, Turok, and their collaborators have argued, implies *finite* temperature and *finite* densities on the branes themselves. But this is a highly tentative conclusion because, so far, no one has been able to get the better of the equations and figure out what would happen should branes slam together. In fact, the analyses so far completed point toward the splat being subject to the same problem that afflicts the inflationary theory at time zero: the mathematics breaks down. Thus, cosmology is still in need of a rigorous resolution of its singular start—be it the true start of the universe, or the start of our current cycle.

The most compelling feature of the cyclic model is the way it incorporates dark energy and the observed accelerated expansion. In 1998, when it was discovered that the universe is undergoing accelerated expansion, it was quite a surprise to most physicists and astronomers. While it can be incorporated into the inflationary cosmological picture by assuming that the universe contains precisely the right amount of dark energy, accelerated expansion seems like a clumsy add-on. In the cyclic model, by contrast, dark energy's role is natural and pivotal. The trillion-year period of slow but steadily accelerated expansion is crucial for wiping the slate clean, for diluting the observable universe to near nothingness, and for resetting conditions in preparation for the next